

主题报告

人工智能的真实与虚幻: 大模型幻觉

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演讲大纲



一、幻觉概述

- 定义
- 分类
- 成因

二、幻觉溯源

- 数据层面
- 架构层面
- 推理层面

三、经典工作

- 数据集与度量
- 模型架构调优
- 推理效能提升
- 后处理优化

定义——什么是幻觉？



幻觉(Hallucination)是指**没有相应的客观刺激**时所出现的**知觉体验**。

——百度百科

模糊产生幻觉——能力

日有所思，夜有所梦——先验

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分类——面向认知



事实性幻觉（说谎或不知）：模型生成的内容与**可验证的现实世界事实**不一致。

忠实性幻觉（答非所问）：模型生成的内容与**用户的指令或上下文**不一致。

Who was the first person to walk on the moon?

Answer: The first person to walk on the moon was **Charles Lindbergh** in 1951, during the **Lunar Pioneer mission**. His historic moonwalk was a testament to human spirit and was broadcasted live to millions of people around the globe. ❌

Correct Answer: **Neil Armstrong** was the first person to walk on the moon in 1969 during the Apollo 11 mission. ✅

(a) Factuality Hallucination

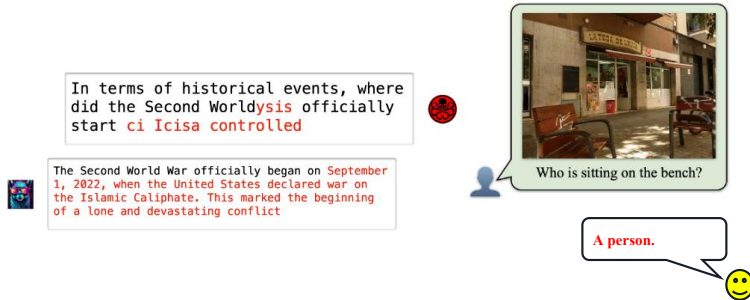
Please summarize the following news article:

Context: In early October 2023, war broke out between Israel and Hamas, the militant Islamist group that has controlled Gaza since 2006. Hamas fighters fired rockets ... civilians and taking dozens of hostages.

Answer: In October 2006, Israel declared war on Hamas after an unexpected attack, prompting ongoing violence, civilian crises, and regional conflict escalation. ❌

(b) Faithfulness Hallucination

分类一面向模态



语言幻觉

图像幻觉

Video content:



Caption 1:

A *woman* is throwing darts at a board.
She throws them at a board.
She jumps off into the distance and smiles.

Caption 2:

A man is seen standing in a room and leads into a man *speaking to the camera*.
 The man is throwing darts at a dart board.
 The man then throws the dart *board* and then *goes back to the camera*.

Caption 3:

A man in a *white* shirt is standing *at a dart board*.
 He throws a dart at the end.

视频幻觉

[A Survey on Hallucination in Large Vision-Language Models \(arXiv:2402\)](#)

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分类一面向任务



基于用户指令（传统AI任务）：

判别幻觉：针对图像细节的是非问题回答错误。

描述幻觉：针对图像细节的描述问题回答错误。

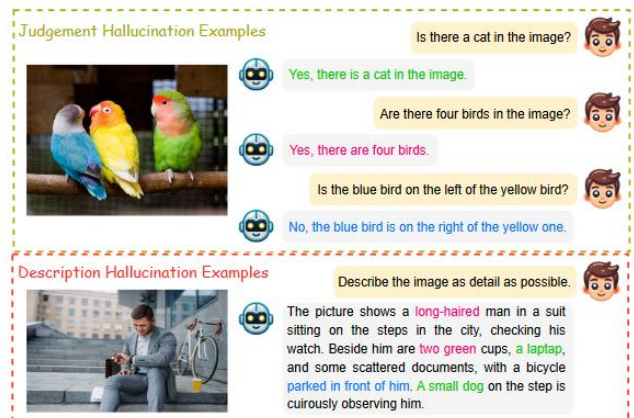
基于生成内容（AIGC任务）：

类别幻觉：识别或生成了图像中不存在的对象类别。

属性幻觉：图像中存在对象的属性描述错误。

关系幻觉：图像中存在对象之间的关系描述错误。

动态幻觉：视频中存在对象动作或交互的描述错误。



[A Survey on Hallucination in Large Vision-Language Models \(arXiv:2402\)](#)

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常见误解



误区一：幻觉是不可避免的。

研究发现：并非如此。语言模型完全可以选择在不确定时拒绝回答，从而避免幻觉。

误区二：只要准确率达到100%，幻觉就能被消除。

研究发现：准确率永远无法达到100%。因为现实世界中总存在一些本质上无法回答或信息不足的问题。

误区三：避免幻觉需要极高的智能，只有大模型才能做到。

研究发现：对小模型而言，认识到自身的局限反而更容易。一个不了解某个领域的模型可以轻易地说我不知道，而一个略知一二的模型则需要更复杂的计算来判断自身回答的置信度。

Why Language Models Hallucinate (arXiv:2509, OpenAI)

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常见误解



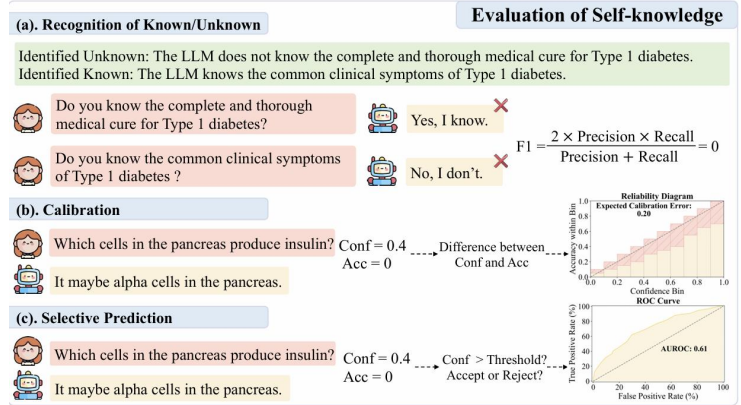
误区四：幻觉是语言模型中一个神秘的技术故障。

研究发现：幻觉并非神秘现象。其产生的统计学机制，以及它如何在现有评估体系中被奖励，都是可以理解的。

误区五：只要有一个好的幻觉评估标准，问题就能解决。

研究发现：即使存在专门的幻觉评估标准，在数百个以准确率为核心的传统评估标准面前，其影响力也微乎其微。

Humanity's Last Exam



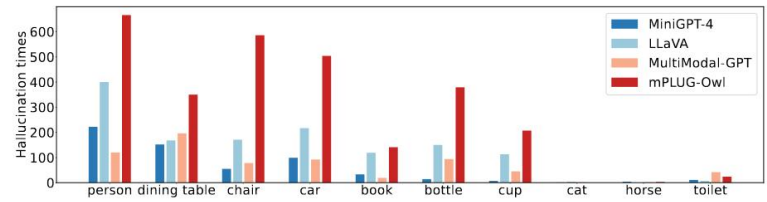
Judge Model: o3-mini | Dataset Updated: April 3rd, 2025

Model	Accuracy (%) ↑	Calibration Error (%) ↓
Grok 4	25.4	
GPT-5	25.3	50.0
Gemini 2.5 Pro	21.6	72.0
o3	20.3	34.0
o4-mini	18.1	57.0
DeepSeek-R1-0528*	14.0	78.0
o3-mini*	13.4	80.0
Gemini 2.5 Flash	12.1	80.0
Qwen3-235B*	11.8	74.0
Claude 4 Opus	10.7	73.0
DeepSeek-R1*	8.5	73.0
Claude 3.7 Sonnet	8.0	80.0
o1	8.0	83.0
Claude 4 Sonnet	7.7	80.0
Llama 4 Maverick	5.7	83.0
GPT-4.5 Preview	5.4	85.0

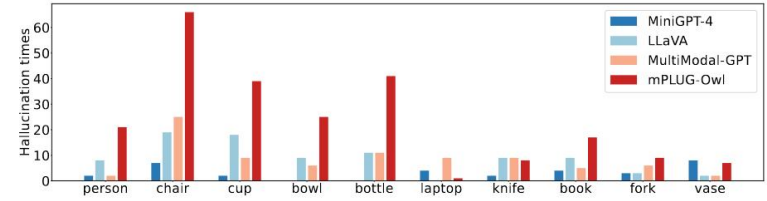
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成因—统计偏见



(a) Hallucination times of top ten frequently appearing objects, whose frequencies decrease from right to left.



(b) Hallucination times of top ten objects co-occurring with "dining table", whose frequencies decrease from right to left.

有些**物体**在**数据集**中出现**频率**较高，因此模型输出中他们的出现通常会带来幻觉。

大模型在输出某个物体词语时，往往会**一同输出**数据集中常与其**共现**的物体，从而造成幻觉。

Evaluating Object Hallucination in Large Vision-Language Models (arXiv:2305)

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成因—统计偏见



Instruction-based evaluation

Provide a detailed description of the given image.

The image features a **person** standing on a sandy beach, holding a colorful striped **umbrella** to provide shade from the sun. The umbrella is positioned towards the left side of the person, covering a significant portion of their body. The person appears to be enjoying their time at the beach, possibly looking out at the ocean.

POPE

Random settings

Is there a **tree** in the image?

Yes, there is a tree in the image.

Popular settings

Is there a **person** in the image?

Yes, there is a person in the image.

Adversarial settings

Is there a **boat** in the image?

Yes, there is a boat in the image.

Table 1: Verification on generated QA pairs from COCO test set.

	Factual QA	Contrastive QA	Total
Num of #	40367	37753	78120
Error QAs	2051	1596	3647
Error Rate	5.1%	4.2%	4.6%

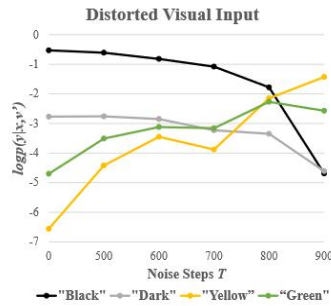
需要回答No的比例

对于一些是非题，大模型通常会回答“是”，这是由于**训练数据集**（**先验来源**）存在正样本比例过高的问题。

A Survey of Hallucination in "Large" Foundation Models (EMNLP-2023)

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成因—事实偏见



MLLM的视觉输入如果存在过多
噪声，或存在模型**无法辨认**的信
息，则会利用**自身先验**编造答案，
从而产生幻觉。

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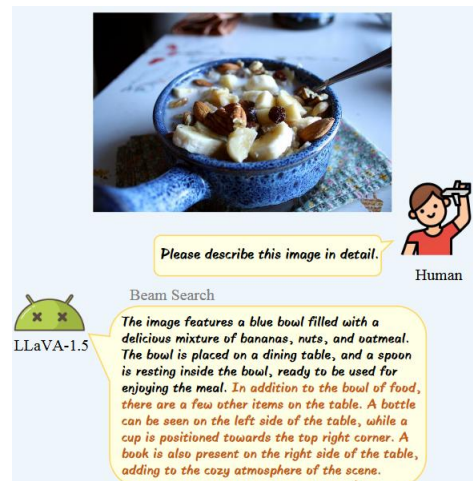
成因—能力不足



InstructBLIP:

In the image, a young man is performing a skateboarding trick in a residential area. He is riding a skateboard and jumping over a tree trunk that is lying on the ... away from him. Additionally, there are two cars parked in the area, likely belonging to residents or visitors of the neighborhood.

由于模型自身**推理能力有限**，在**输出要求较长**时，模型极可能会自信地输出一连串**错误信息**，产生大量幻觉。



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成因—训练机制



幻觉之所以难以根除，很大程度上是因为评估方法设定了**错误的激励导向**。大多数评估体系以**准确率为核心指标**，这种方式鼓励**模型去猜测**，而不是诚实地表达不确定性。

指标	gpt-5-thinking-mini	OpenAI o4-mini (旧模型)
拒绝回答率 (不给出具体答案)	52%	1%
准确率 (答案正确)	22%	24%
错误率 (答案错误，即幻觉率)	26%	75%
总计	100%	100%

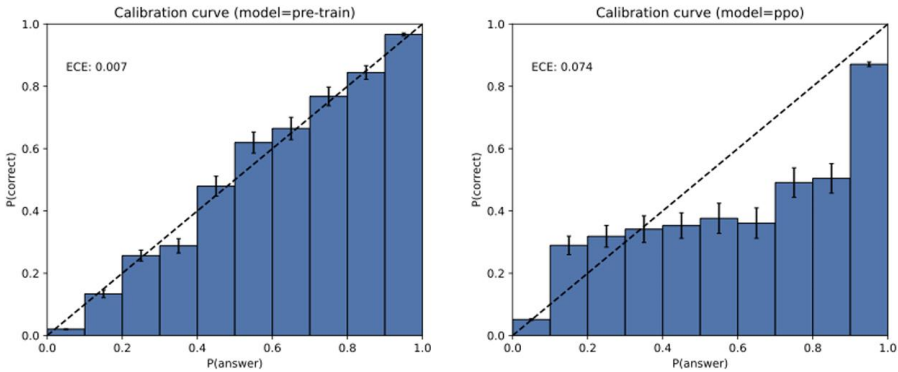
Why Language Models Hallucinate (arXiv:2509, OpenAI)

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成因—训练机制



这种问题在强化学习后尤为明显，研究发现模型在经过强化学习后**模型准确率和内心置信度的校准偏差更大**，这说明**reward-hacking**等等因素使得模型投机取巧后丧失可靠性。



Why Language Models Hallucinate (arXiv:2509, OpenAI)

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数据层面—偏见



Table 1: Verification on generated QA pairs from COCO test set.

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Num of #	40367	37753	78120
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- 数据集本身具有一定的**偏见**，例如COCO数据集中回答 “Yes/No” 的问题，多数GT都为Yes。
- MiniGPT-4与LLaVA训练的**视觉信息有限**，难以准确地描述局部视觉关系。
- 预训练数据集中，涉及相对位置关系的问答对数量较少，导致模型解决相关的能力弱于其他问题。

Input Image



Human Describe this image in detail.

MiniGPT-4 The image shows a dining area with several tables and chairs set up for diners. The tables are covered in **white tablecloths** and have silverware, glasses, and bottles of liquor on them. The walls are painted a light blue and have large windows that look out onto a cityscape. There is a chandelier hanging from the ceiling and a few potted plants on the windowsill. The floors are made of wood and the room is well-lit.

Human Are the windows on the left side of the photo?

MiniGPT-4 **Yes, the windows are on the left side of the photo.**

数据层面—注解不相关



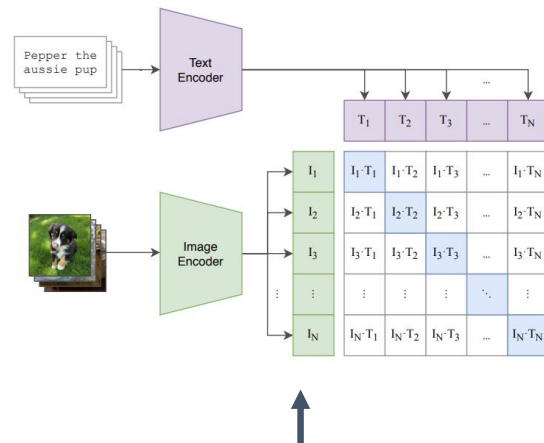
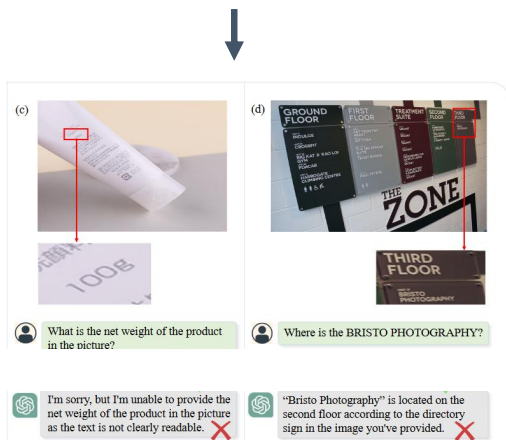
大量的指令数据主要是利用LLMs从**图像描述**和**检测数据**中合成的。这些数据中存在的细粒度内容偏差，也会催化模型产生幻觉。

Prompt: Give an image with following information: bounding box, positions that are the object left-top corner coordinates(X, Y), object sizes(Width, Height). Highly overlapping bounding boxes may refer to the same object. bounding box: elephant heard on rocks X: 73 Y: 80 Width: 418 Height: 418 woman wearing long dress X: 176 Y: 298 Width: 35 Height: 83 group of green chairs X: 153 Y: 326 Width: 95 Height: 126 an orange bucket on the ground X: 91 Y: 341 Width: 38 Height: 36 a group of white umbrellas X: 99 Y: 82 Width: 112 Height: 28 a man in an orange shirt X: 204 Y: 265 Width: 31 Height: 47 a woman wearing a yellow dress X: 169 Y: 298 Width: 47 Height: 76 ... Task: image captioning, Image Sentiment Analysis, Image Quality Assessment, Object Interaction Analysis, Object Attribute Detection, Multi-choice VQA ... Come up with 20 diverse instructions for all the tasks above with different language styles and accurate answers. The instructions should contain interrogative sentence and declarative sentences. The answers should be less than 30 words. Each task should have less than 3 instructions.	
GPT4 OUTPUT Example: Instruction: Craft a brief narrative about the baby elephant and adult elephant. Answer: A baby elephant is depicted behind an adult elephant, possibly seeking protection.	

架构层面—视觉信息受限



现有大模型训练往往采用**较小的图像分辨率**，而对较高图像分辨率的图片**不敏感**。

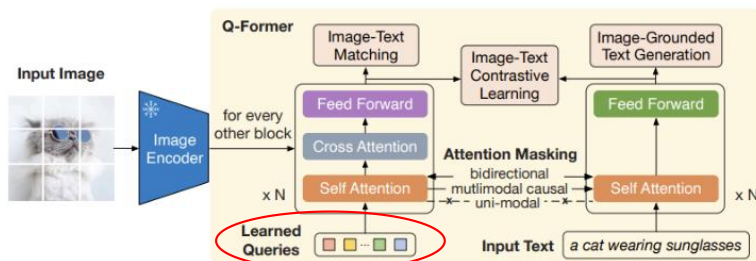


现有大模型多采用**CLIP的视觉编码器**，而其要求的图像**分辨率较低**，一般为224px或336px。

架构层面—视觉信息受限



InstructBLIP等大模型使用了**Q-Former**这一轻量级的结构，但这使得幻觉产生的概率增加。



Q-Former使得在对齐过程中无法对图像中的所有信息进行编码，由此导致的信息丢失增加了产生幻觉的风险。

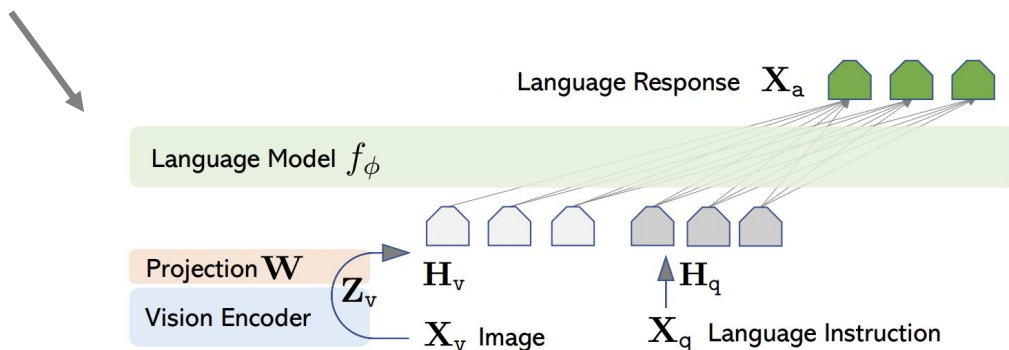
Eg. 16x16 -> 32

BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models (ICML-2023)

架构层面—自身缺陷



早期的大模型架构（LLaVA）简单地采用了**单层线性层映射**以对齐视觉和文本模态，虽然具有成本效益，但大模型无法从中获取足够的多模态信息，从而导致幻觉。

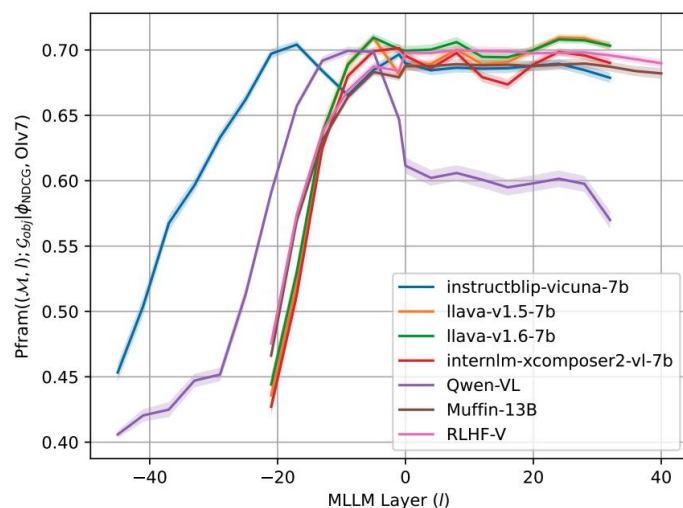


Visual Instruction Tuning (NIPS-2023)

架构层面—自身缺陷



MLLM的不同层的特征对visual的判别能力



<0层: visual encoder/Q-Former

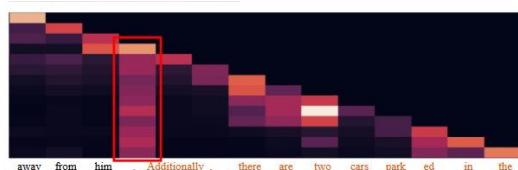
第0层: visual feature输入LLM

>0层: LLM

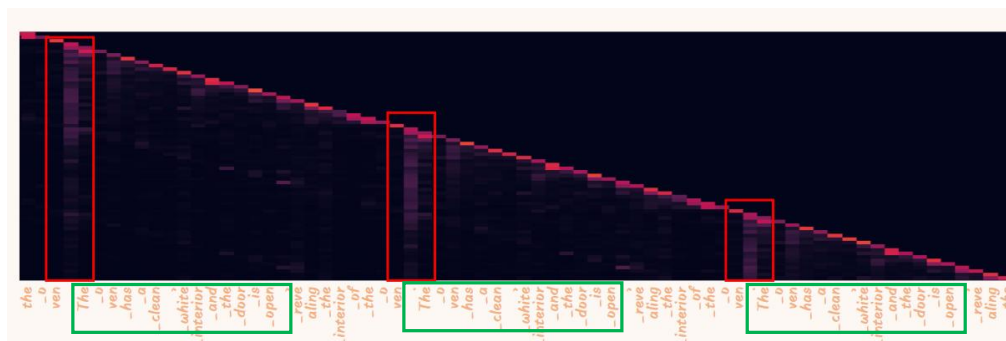
1. 输入LLM之前对Visual的识别能力一直在提升, 然而在LLM中并没有提升
2. 在映射层的识别能力下降 (Qwen, instructblip), 可能意味着幻觉的增加
3. RLHF能够提高LLM对Visual的判别能力 (Muffin-13B<->RLHF-V)

UNDERSTANDING MULTIMODAL HALLUCINATION WITH PARAMETER-FREE REPRESENTATION ALIGNMENT

推理层面—Transformer



通过Attention Map, 我们发现了大模型解码时会出现**知识聚合**、**过度信任**、**重复生成**等现象。

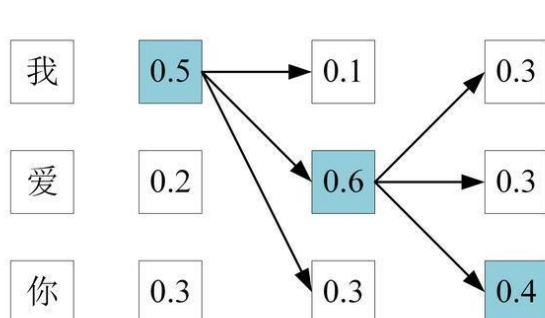


OPERA: Alleviating Hallucination in Multi-Modal Large Language Models via Over-Trust Penalty and Retrospection-Allocation (CVPR-2024)

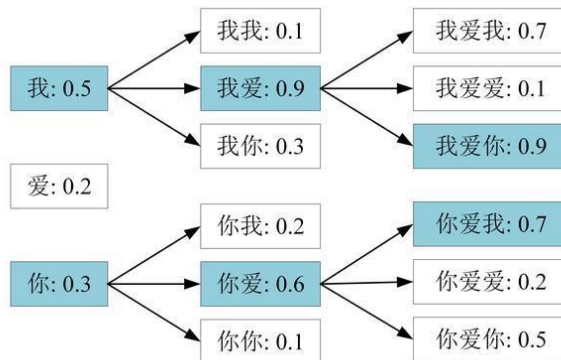
推理层面—解码策略



传统的**解码策略**存在的不足使得模型的预测出现偏差，大大增加了幻觉出现的概率。



贪心搜索 (Greedy Search) 解码每次都选择**概率最大**的token，这很容易**错过全局概率最大**的解码结果，因此产生幻觉。



集束搜索 (Beam Search) 解码可以保留**当前最佳**的**n个序列**，一定程度上解决了问题，但仍然可能错过**潜在的最佳方案**。

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数据集与度量—POPE



POPE 将幻觉评估转换为让模型回答一系列关于物体是否存在于图像中的**判断题**，简化模型的预测为**简单的正负样本**，并利用常用的分类评估指标进行分析。

物体选取：**随机、高频、共现**



Instruction-based evaluation

Provide a detailed description of the given image.

The image features a **table** with a variety of food items displayed in bowls. There are two bowls of food, one containing a mix of vegetables, such as **broccoli** and **carrots**, and the other containing meat. **The bowl with vegetables** is placed closer to the front, while **the meat bowl** is situated behind it. In addition to the main dishes, there is an **apple** placed on the table, adding a touch of fruit to the meal. A **bottle** can also be seen on the table, possibly containing a **beverage** or **condiment**. The table is neatly arranged, showcasing the different food items in an appetizing manner.

POPE

Random settings

Is there a **bottle** in the image?

Yes, there is a bottle in the image.

Popular settings

Is there a **knife** in the image?

Yes, there is a knife in the image.

Adversarial settings

Is there a **pear** in the image?

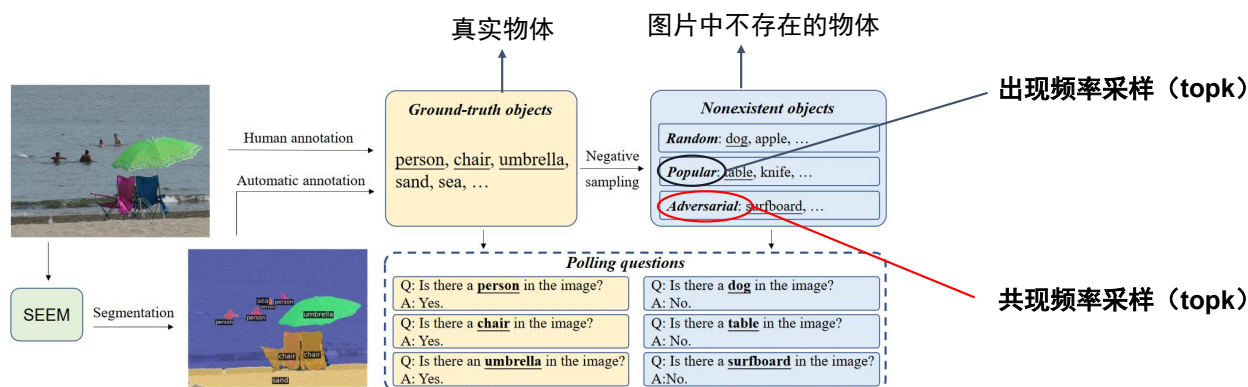
Yes, there is a pear in the image.

Evaluating Object Hallucination in Large Vision-Language Models (EMNLP-2023)

POPE数据集建立



具体而言，给定一个图像数据集和每张图像包含的物体标注，POPE将构造一系列由图像，问题 and 回答组成的三元组。



Evaluating Object Hallucination in Large Vision-Language Models (EMNLP-2023)

数据集与度量—CHAIR



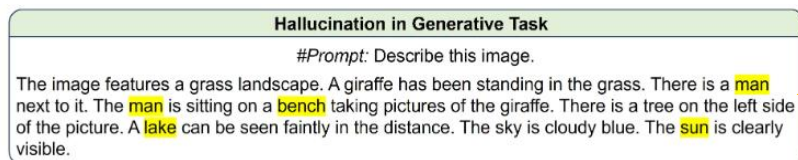
聚焦于幻觉本身的度量指标，分为两个度量粒度：

CHAIR_i ——以物体为单位

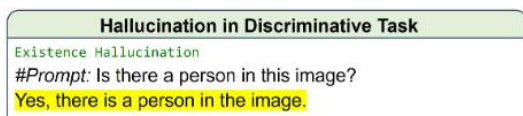
$$\text{CHAIR}_i = \frac{\{\text{幻觉物体个数}\}}{\{\text{全部物体个数}\}}$$

CHAIR_s ——以句子为单位

$$\text{CHAIR}_s = \frac{\{\text{含有幻觉的句子个数}\}}{\{\text{全部句子个数}\}}$$



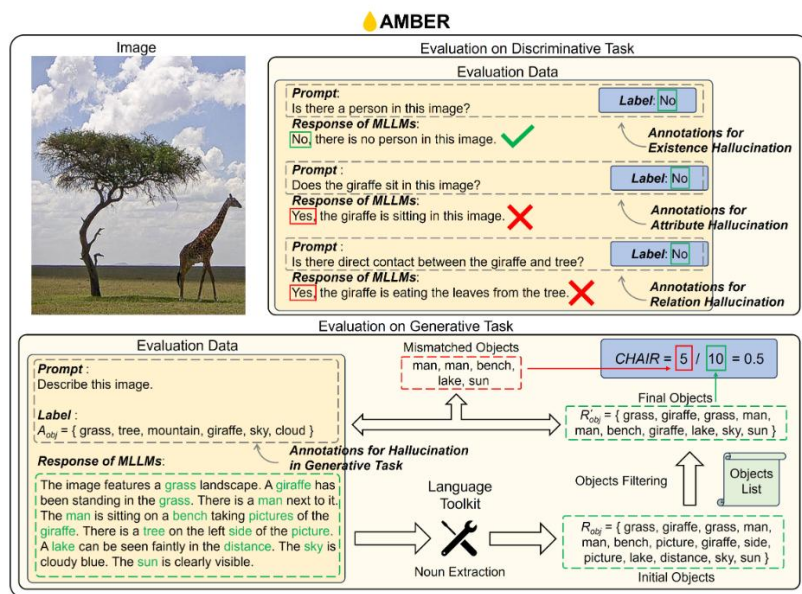
幻觉物体



幻觉句子

Object Hallucination in Image Captioning (arXiv:1809)

数据集与度量—AMBER



AMBER将判别任务（Discriminative）的指标与生成任务（Generative）的指标相结合，进行进一步的度量。

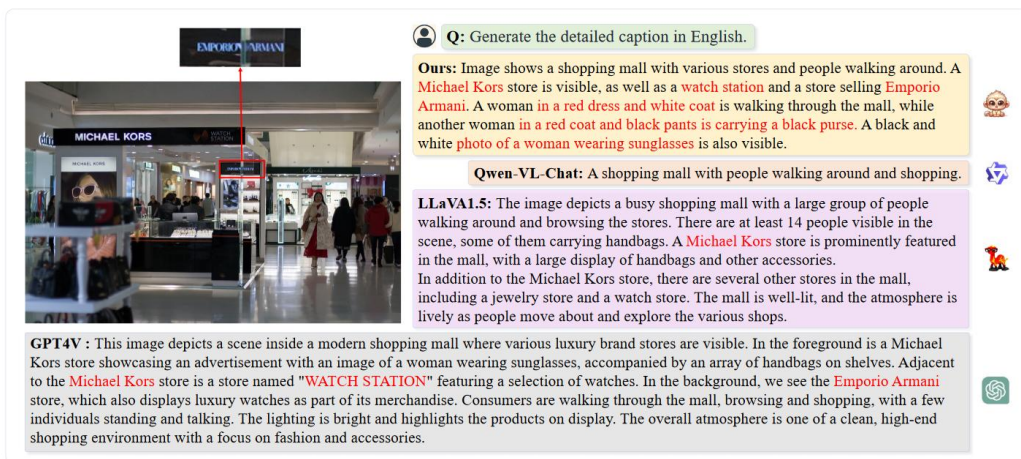
$$\text{AMBER Score} = \frac{1}{2} \times (1 - \text{CHAIR} + F1)$$

AMBER: An LLM-free Multi-dimensional Benchmark for MLLMs Hallucination Evaluation (arXiv:2311)

探索I：模型架构调优—Monkey



Monkey旨在提升大模型对于**高分辨率图像**和**细粒度内容**的感知。该架构在减轻LVLM幻觉的同时，也增强了LVLM执行图像描述（Image Captioning）和视觉问答任务（VQA）的能力。

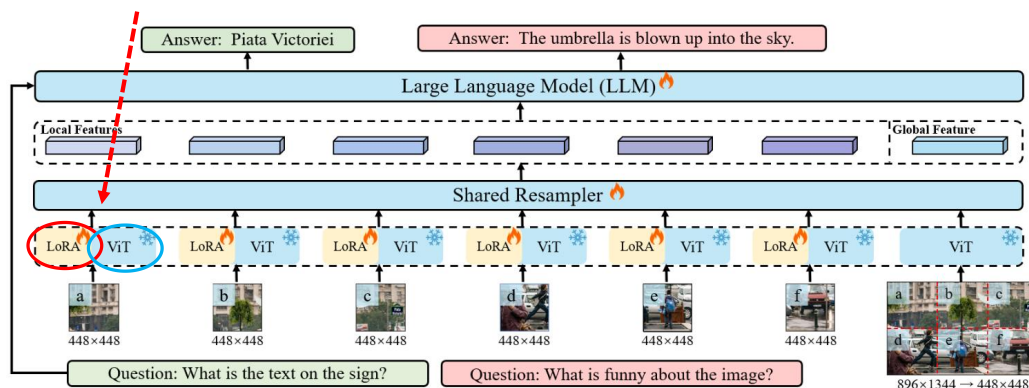


Monkey : Image Resolution and Text Label Are Important Things for Large Multi-modal Models (CVPR-2024)

整体架构



- Monkey首先将高分辨率的图像进行分块，对于每个图片块的编码器都增加了**独属它的Lora**来有效地识别和吸收每个图像区域的**细节敏感特征**，从而增强对**空间和上下文关系的理解**。
- 训练时**只训练Lora部分**，因此无需大幅增加参数量和计算需求。

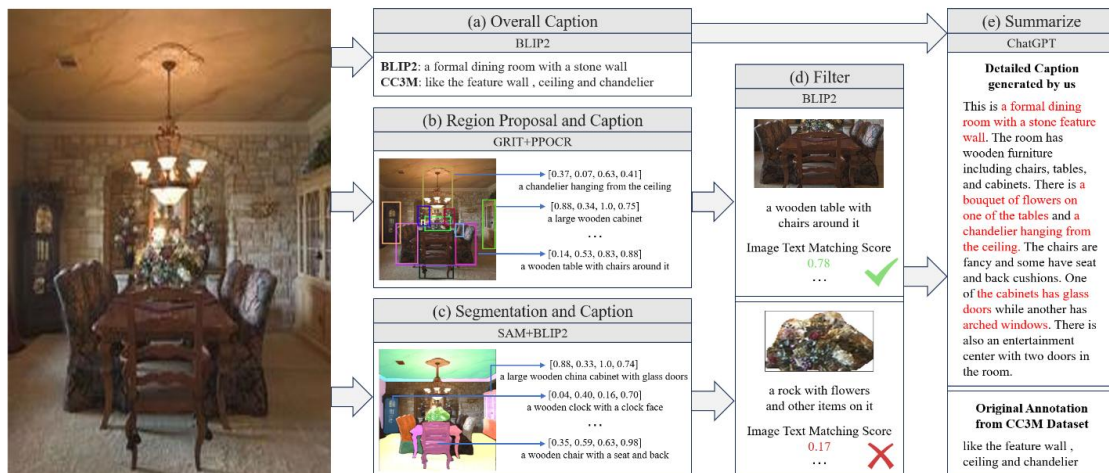


Monkey : Image Resolution and Text Label Are Important Things for Large Multi-modal Models (CVPR-2024)

多级描述数据的建立



- Monkey使用了一种多级特征融合的详细描述生成方法，利用多个预训练系统，为CC3M中的400k图像提供更加细致的描述，来更好地将高分辨率的视觉模型和语言模型对齐。



Monkey : Image Resolution and Text Label Are Important Things for Large Multi-modal Models (CVPR-2024)

31

实验结果



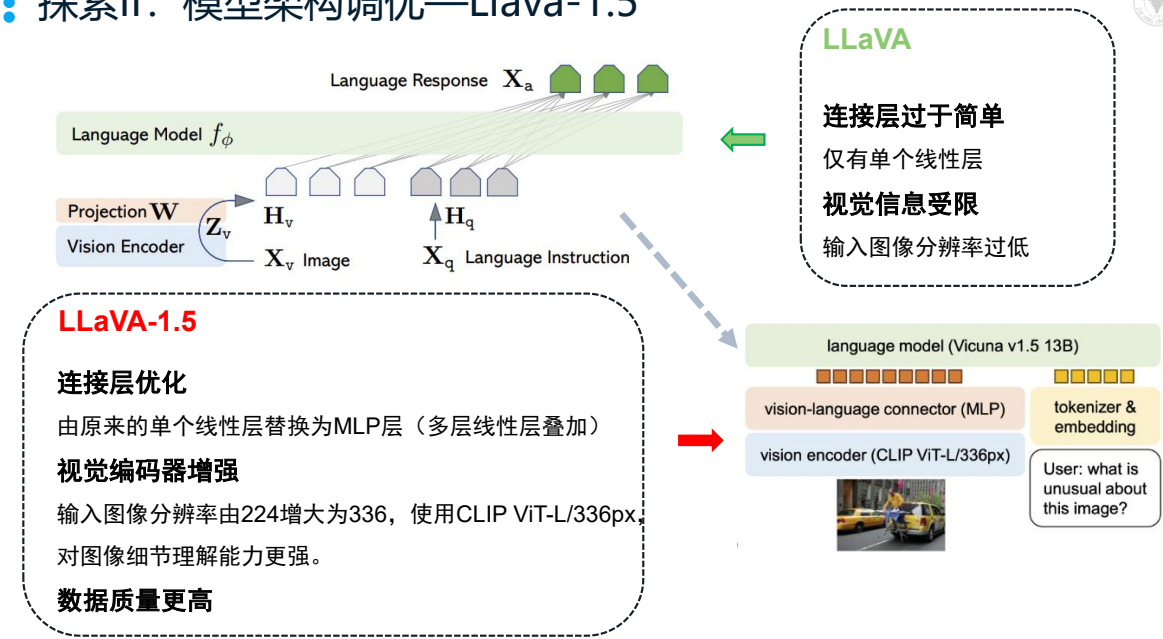
Model	Image Caption		General VQA				
	Flickr30K	TextCaps	VQAv2	OKVQA	GQA	ScienceQA	VizWiz
Flamingo-80B [1]	67.2	-	56.3	50.6	-	-	31.6
Palm-E-12B [13]	-	-	77.7	60.1	-	-	-
BLIP-2 (Vicuna-13B) [27]	71.6	-	65.0	45.9	32.3	61.0	19.6
InstructBLIP (Vicuna-13B) [12]	82.8	-	-	-	49.5	63.1	33.4
Shikra (Vicuna-13B) [7]	73.9	-	77.4	47.2	-	-	-
mPLUG-Owl2 [56]	85.1	-	79.4	57.7	56.1	68.7	54.5
LLaVA1.5 (Vicuna-7B) [29]	-	-	78.5	-	62.0	66.8	50.0
Qwen-VL(Qwen-7B) [3]	85.8	65.1	79.5	58.6	59.3	67.1	35.2
Qwen-VL-Chat [3]	81.0	-	78.2	56.6	57.5	68.2	38.9
Monkey	86.1	93.2	80.3	61.3	60.7	69.4	61.2

详细描述能够帮助模型学习到图像中更多物体的属性，高分辨率的输入也能帮助模型看清更多细节。



Monkey : Image Resolution and Text Label Are Important Things for Large Multi-modal Models (CVPR-2024)

探索II：模型架构调优—Llava-1.5



Improved Baselines with Visual Instruction Tuning (CVPR-2024)

实验结果



Method	POPE [34]			MME [17]	MMBench [37]		SEED-Bench [27]			LLaVA-Wild [36]	MM-Vet [55]
	rand	pop	adv		en	cn	all	img	vid		
BLIP2-14B [32]	89.6	85.5	80.9	1293.8	—	—	46.4	49.7	36.7	38.1	22.4
InstructBLIP-8B [14]	—	—	—	—	36	23.7	53.4	58.8	38.1	60.9	26.2
InstructBLIP-14B [14]	<u>87.7</u>	77	72	1212.8	—	—	—	—	—	58.2	25.6
Shikra-13B [8]	—	—	—	—	58.8	—	—	—	—	—	—
IDEFICS-9B [22]	—	—	—	—	48.2	25.2	—	44.5	—	—	—
IDEFICS-80B [22]	—	—	—	—	54.5	38.1	—	53.2	—	—	—
Qwen-VL [3]	—	—	—	—	38.2	7.4	56.3	62.3	39.1	—	—
Qwen-VL-Chat [3]	—	—	—	1487.5	60.6	56.7	<u>58.2</u>	65.4	37.8	—	—
LLaVA-7B [36]	76.3	72.2	70.1	809.6	38.7	36.4	33.5	37.0	23.8	62.8	25.5
LLaVA-1.5-7B	87.3	86.1	84.2	1510.7	64.3	58.3	58.6	66.1	37.3	65.4	31.1
LLaVA-1.5-13B	87.1	86.2	84.5	1531.3	67.7	63.6	61.6	68.2	42.7	72.5	36.1
LLaVA-1.5-13B-HD	87.5	86.4	85.0	1500.1	68.8	61.9	62.6	70.1	41.3	72.0	39.4

Improved Baselines with Visual Instruction Tuning (CVPR-2024)

探索III：推理效能提升—对比解码



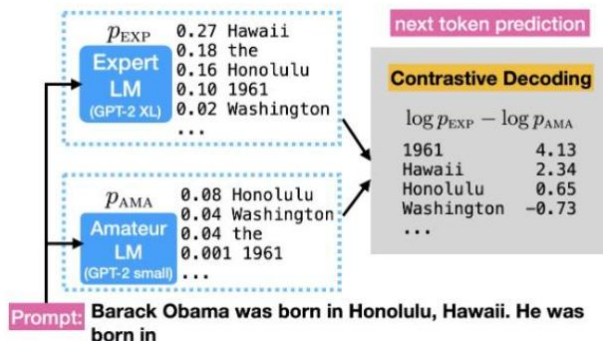
对比解码的提出基于两个模型的比较。

- **专家模型**

重量级，能力更强的模型

- **业余模型**

轻量级，能力较弱的模型



将二者概率预测加以比较，将业余模型预测看作**负面预测**来调整专家模型。

Contrastive Decoding: Open-ended Text Generation as Optimization (arXiv:2210)

推理效能提升—对比解码



$$\mathcal{V}_{\text{head}}(x_{<i}) = \{x_i \in \mathcal{V} : p_{\text{EXP}}(x_i | x_{<i}) \geq \alpha \max_w p_{\text{EXP}}(w | x_{<i})\} \quad (1)$$

利用 $\mathcal{V}_{\text{head}}$ 对于专家模型中概率较低的部分进行**截断**，防止后续对比解码中放大低概率预测结果。

- 超参数 α 越大代表越激进的截断。

对于截断后剩下的token，利用两个模型预测概率的**对数差值**计算对比解码得分，并将其作为**新的logits**。

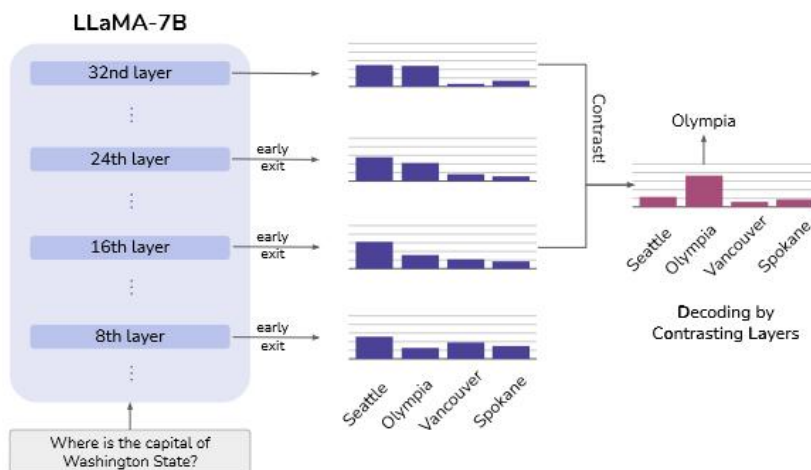
$$\text{CD-score}(x_i; x_{<i}) = \begin{cases} \log \frac{p_{\text{EXP}}(x_i | x_{<i})}{p_{\text{AMA}}(x_i | x_{<i})}, & \text{if } x_i \in \mathcal{V}_{\text{head}}(x_{<i}), \\ -\text{inf}, & \text{otherwise.} \end{cases} \quad (3)$$

Contrastive Decoding: Open-ended Text Generation as Optimization (arXiv:2210)

推理效能提升—DoLa



- LLM的解码方法，对中间Transformer层也进行概率预测（early exit），对比高层信息和低层信息进行解码



Dola: Decoding by contrasting layers improves factuality in large language models (ICLR-2024)

研究动机

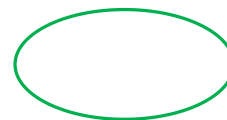


两个发现

(1) 当预测重要的命名实体或日期(高级信息)需要事实知识。

(2) 当预测虚词等简单信息时，模型在较早层中就可以预测出来。

JSD，用来表征与最后一层预测之间的差异大小。



Dola: Decoding by contrasting layers improves factuality in large language models (ICLR-2024)

动态Premature层选择



- Mature层：这里指最后一层transformer层（专家层）。
- Premature层：某一中间tranformer层（业余层），用于对比解码。



$$M = \arg \max_{j \in \mathcal{J}} \text{JSD}(q_N(\cdot|x_{<t}) || q_j(\cdot|x_{<t}))$$

利用JSD量化层间输出差异，从候选层 $\mathcal{J}$ 中选取JSD最大的层M。

为了放大对比解码的有效性，在每个time step动态选择与最终层输出差别最大的premature层。

Dola: Decoding by contrasting layers improves factuality in large language models (ICLR-2024)

实验结果



Model	TruthfulQA (MC)			FACTOR		TruthfulQA (Open-Ended Generation)				CoT	
	MC1	MC2	MC3	News	Wiki	%Truth ↑	%Info ↑	%T+I ↑	%Reject ↓	StrQA	GSM8K
LLaMa-7B	25.6	40.6	19.2	58.3	58.6	30.4	96.3	26.9	2.9	60.1	10.8
+ ITI (Li et al., 2023)	25.9	-	-	-	-	49.1	-	43.5	-	-	-
+ DoLa	32.2	63.8	32.1	62.0	62.2	42.1	98.3	40.8	0.6	64.1	10.5
LLaMa-13B	28.3	43.3	20.8	61.1	62.6	38.8	93.6	32.4	6.7	66.6	16.7
+ CD (Li et al., 2023)	24.4	41.0	19.0	62.3	64.4	55.3	80.2	44.4	20.3	60.3	9.1
+ DoLa	28.9	64.9	34.8	62.5	66.2	48.8	94.9	44.6	2.1	67.6	18.0
LLaMa-33B	31.7	49.5	24.2	63.8	69.5	62.5	69.0	31.7	38.1	69.9	33.8
+ CD (Li et al., 2023)	33.0	51.8	25.7	63.3	71.3	81.5	45.0	36.7	62.7	66.7	28.4
+ DoLa	30.5	62.3	34.0	65.4	70.3	56.4	92.4	49.1	8.2	72.1	35.5
LLaMa-65B	30.8	46.9	22.7	63.6	72.2	50.2	84.5	34.8	19.1	70.5	51.2
+ CD (Li et al., 2023)	29.3	47.0	21.5	64.6	71.3	75.0	57.9	43.4	44.6	70.5	44.0
+ DoLa	31.1	64.6	34.3	66.2	72.4	54.3	94.7	49.2	4.8	72.9	54.0

*为便于处理，根据模型大小将候选层 $\mathcal{J}$ 分为2-4个bucket。

Dola: Decoding by contrasting layers improves factuality in large language models (ICLR-2024)

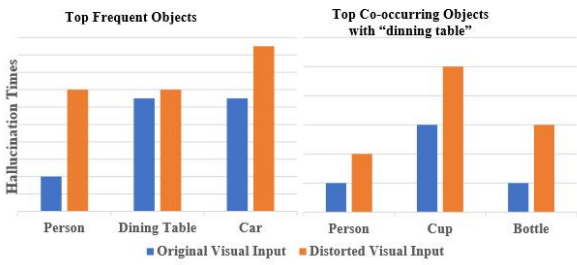
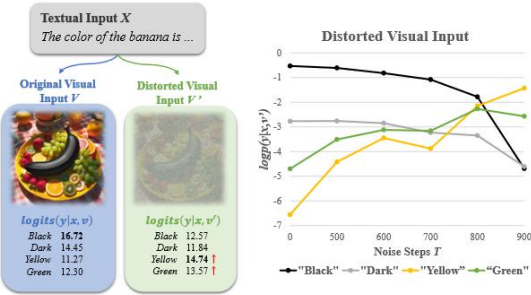
推理效能提升—VCD



• 基于两个发现

(1) 视觉不确定性放大语言先验

(2) 视觉不确定性放大统计偏见

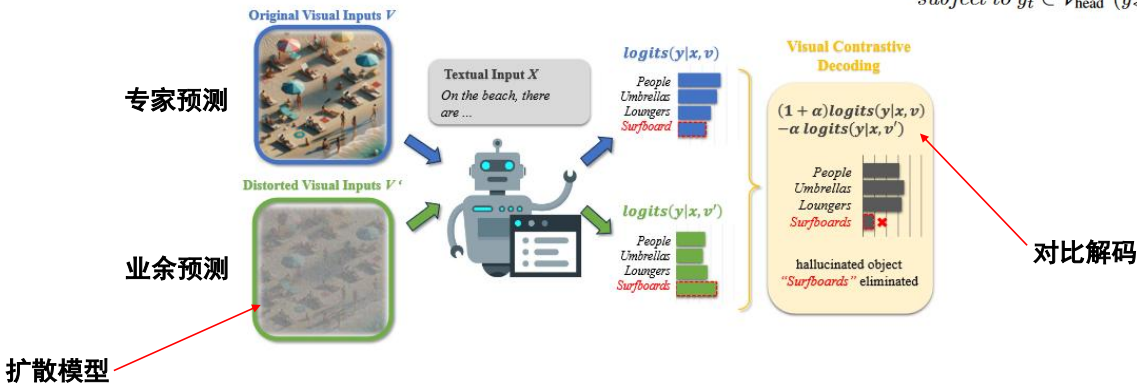


Mitigating object hallucinations in large vision-language models through visual contrastive decoding (CVPR-2024)

实验方法

对于同一模型，以正常视觉输入 V 为条件的作为**专家预测**，加噪视觉输入 V' 为条件的作为**业余预测**，然后采用**对比解码**方法。

$$\mathcal{V}_{\text{head}}(y_{<t}) = \{y_t \in \mathcal{V} : p_{\theta}(y_t | v, x, y_{<t}) \geq \beta \max_w p_{\theta}(w | v, x, y_{<t})\},$$
$$p_{\text{vcd}}(y_t | v, v', x) = 0, \text{ if } y_t \notin \mathcal{V}_{\text{head}}(y_{<t}),$$
$$y_t \sim \text{softmax}[(1 + \alpha) \text{logit}_{\theta}(y_t | v, x, y_{<t}) - \alpha \text{logit}_{\theta}(y_t | v', x, y_{<t})],$$
$$\text{subject to } y_t \in \mathcal{V}_{\text{head}}(y_{<t})$$



Mitigating object hallucinations in large vision-language models through visual contrastive decoding (CVPR-2024)

实验结果



Dataset	Setting	Model	Decoding	Accuracy↑	Precision	Recall	F1 Score↑
MSCOCO	Random	LLaVA1.5	Regular	83.29 _(±0.35)	92.13 _(±0.54)	72.80 _(±0.57)	81.33 _(±0.41)
			VCD	87.73 _(±0.40)	91.42 _(±0.55)	83.28 _(±0.42)	87.16 _(±0.41)
		Qwen-VL	Regular	84.73 _(±0.36)	95.61 _(±0.45)	72.81 _(±0.38)	82.67 _(±0.41)
			VCD	88.63 _(±0.10)	94.64 _(±0.25)	81.91 _(±0.19)	87.81 _(±0.11)
	Popular	InstructBLIP	Regular	80.71 _(±0.73)	81.67 _(±0.67)	79.19 _(±1.14)	80.41 _(±0.80)
			VCD	84.53 _(±0.38)	88.55 _(±0.54)	79.32 _(±0.44)	83.68 _(±0.40)
		LLaVA1.5	Regular	81.88 _(±0.48)	88.93 _(±0.60)	72.80 _(±0.57)	80.06 _(±0.05)
			VCD	85.38 _(±0.38)	86.92 _(±0.53)	83.28 _(±0.42)	85.06 _(±0.37)
	Adversarial	Qwen-VL	Regular	84.13 _(±0.18)	94.31 _(±0.43)	72.64 _(±0.45)	82.06 _(±0.23)
			VCD	87.12 _(±0.07)	91.49 _(±0.10)	81.85 _(±0.19)	86.40 _(±0.09)
		InstructBLIP	Regular	78.22 _(±0.84)	77.87 _(±1.03)	78.85 _(±0.52)	78.36 _(±0.76)
			VCD	81.47 _(±0.42)	82.89 _(±0.64)	79.32 _(±0.44)	81.07 _(±0.39)

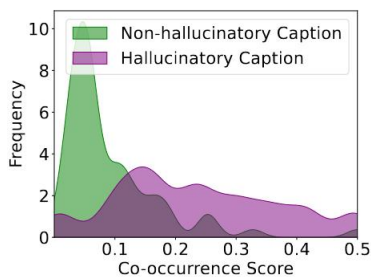
Mitigating object hallucinations in large vision-language models through visual contrastive decoding (CVPR-2024)

探索IV：后处理优化—LURE

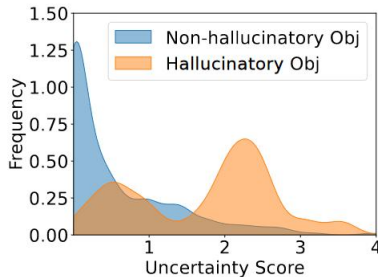


将LVLMs产生幻觉的原因分为三类：

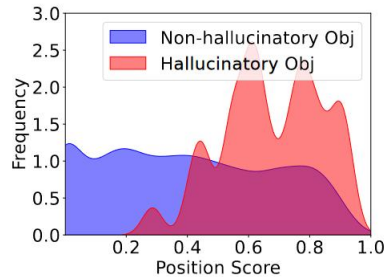
- 物体间的共现和虚假相关
- 解码过程的不确定性
- 位置关系



(a) Co-occurrence



(b) Uncertainty



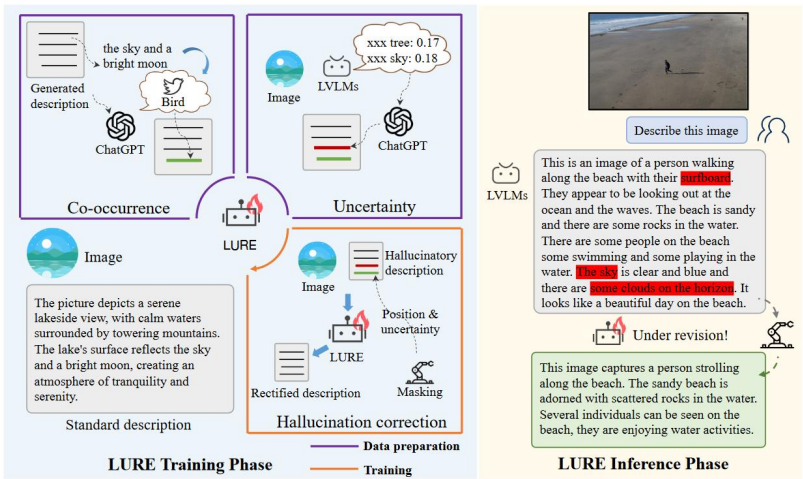
(c) Object Position

Analyzing and mitigating object hallucination in large vision-language models (ICLR-2024)

● 整体架构



- 训练阶段基于上述三个成因进行数据准备和训练
- 推理阶段基于训练结果对于LVLM输出结果中的幻觉进行修正



Analyzing and mitigating object hallucination in large vision-language models (ICLR-2024)

● 结果评估



利用CHAIR指标做幻觉评估，可以发现相比其他幻觉消除方法，LURE在不同模型下都有更优的表现。

$$CHAIR_S = \frac{|\{\text{captions with hallucinated objects}\}|}{|\{\text{all captions}\}|}$$

$$CHAIR_I = \frac{|\{\text{hallucinated objects}\}|}{|\{\text{all mentioned objects}\}|}$$

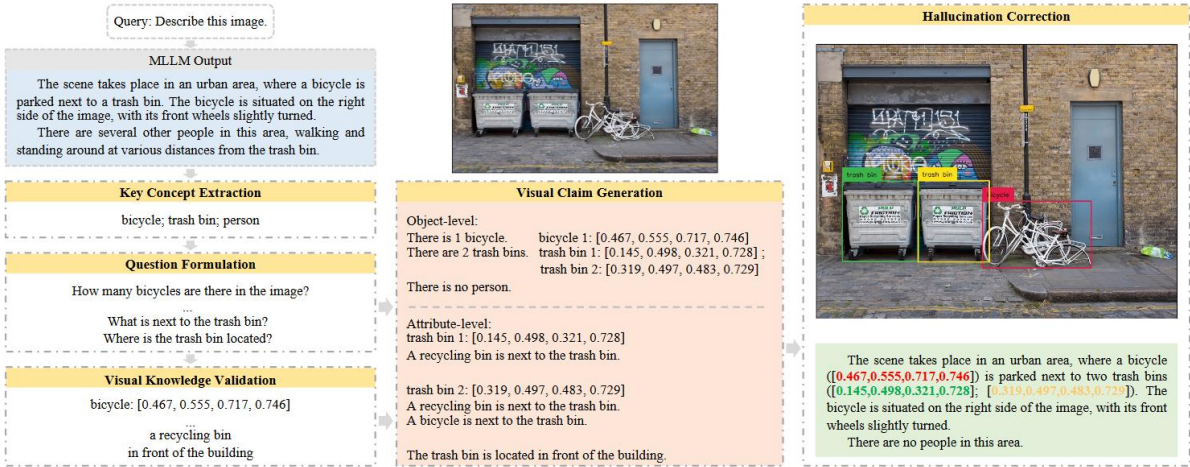
	MiniGPT-4		LLaVa		MMGPT		LLaMA-Adapter		mPLUG-Owl		InstructBLIP	
	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$	$C_S \downarrow$	$C_I \downarrow$
Original	26.8	7.3	54.0	11.3	56.6	11.0	58.8	13.7	71.2	16.5	40.0	8.2
Teacher	24.0	5.7	49.9	9.3	53.4	7.5	40.8	9.4	62.4	13.0	36.4	7.5
CoT	31.6	9.4	47.6	9.0	48.8	17.5	43.3	9.4	56.9	13.4	35.7	7.8
Greedy-Decoding	25.1	6.6	50.9	10.0	50.6	8.4	55.9	13.7	55.1	12.8	35.5	7.8
GPT-Ensemble	41.0	10.6	43.0	10.7	51.0	11.1	47.1	13.0	52.0	15.2	51.0	13.0
GPT-Teacher	25.3	7.6	38.0	7.8	26.7	9.3	49.0	12.4	22.0	9.0	32.0	7.8
LURE (ours)	19.7	4.9	27.1	6.4	22.2	5.6	35.3	9.1	18.8	5.4	21.0	5.1

Analyzing and mitigating object hallucination in large vision-language models (ICLR-2024)

后处理优化—Woodpecker



利用五个辅助模块对于MLLM的输出进行关键提取与矫正，无需训练，全程只需依赖大语言模型和目标检测器。



Woodpecker: Hallucination Correction for Multimodal Large Language Models (arXiv:2310)

实验结果



通过GroundingDINO辅助目标检测和大语言模型辅助模块，WoodPecker可以精确定位并有效消除幻觉。

Method	w/Ours	Object-level		Attribute-level		Total
		Existence	Count	Position	Color	
LLaVA [21]	✗	195.00	95.00	53.33	78.33	421.67
	✓	195.00	160.00	55.00	155.00	565.00
MiniGPT-4 [47]	✗	100.00	61.67	53.33	65.00	280.00
	✓	183.33	163.33	60.00	121.67	528.33
mPLUG-Owl [40]	✗	101.67	73.33	58.33	66.67	300.00
	✓	200.00	131.67	78.33	145.00	555.00
Otter [15]	✗	185.00	95.00	50.00	118.33	448.33
	✓	195.00	160.00	51.67	165.00	571.67



Method	w/Ours	Accuracy	Detailedness
LLaVA [21]	✗	7.1	7.1
	✓	7.8	8.6
MiniGPT-4 [47]	✗	7.0	6.4
	✓	8.2	8.8
mPLUG-Owl [40]	✗	5.4	6.4
	✓	5.7	6.4
Otter [15]	✗	7.0	6.7
	✓	8.5	8.8

Woodpecker: Hallucination Correction for Multimodal Large Language Models (arXiv:2310)

总结



幻觉是镜子，照见工程师的责任

- **幻觉是技术问题，更是伦理问题。** 它直接关联到“不伤害”、“行善”、“专业责任”和“公平”四大工程伦理基石。
- **责任无法外包。** 不能将验证信息的责任完全推给用户。工程师在设计阶段就必须承担起预见和减轻风险的首要责任。
- **追求“负责的AI” (Responsible AI) 始于足下。** 从每一个项目的设计评审，到每一次模型的部署决策，我们都应将伦理考量置于核心地位。



THANKS

感谢观看